PAGE No.
DATE / /

4/8/26

Machine Learning
Assignment - 37

1) Define Artificial Intelligence (AI). How is AI different from traditional software systems?

Ans: Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence, such as learning, reasoning, problem-solving, decision-making, and understanding language.

In simple words:
AI is the ability of a computer or machine to think, learn from data, and make ~~it~~ intelligent decisions like a human.

Traditional software systems work based on fixed rules written by the programmer. They always produce the same output for the same input and cannot learn from experience. AI systems, on the other hand, learn from data and improve their performance over time. Instead of depending only on predefined rules, they identify patterns and make predictions or decisions.

Example -
A traditional calculator always follows mathematical formulas written by the programmer.

A voice assistant like Siri or Google Assistant learns speech patterns and understands different ways of asking the same questions.

2) Explain the three types of AI based on capability:

- Narrow AI -
Narrow AI is ~~degr~~ designed to perform only one specific task. It cannot perform tasks outside the area for which it was trained.
Examples - Google Assistant, Siri, Face Recognition, Chatbots

- General AI -
General AI refers to machine that can perform any intellectual task that a human can do. It can learn, understand, reason, and solve different kinds of problems without being trained seperatly for each task.
At present, General AI does not exist. It still a research goal.
Example - A robot that can study, drive a ~~care~~ car, cook food.

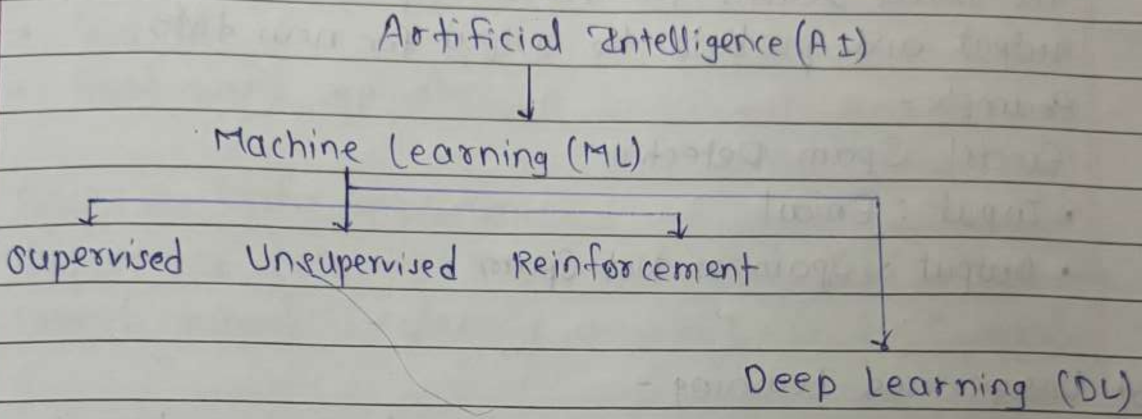
- Super AI -
Super AI is a hypothetical type of AI that would be more intelligent that humans in every field, including reasoning, creativity, problem-solving, and decision-making.
Currently, Super AI does not exist and is only a theoretical concept

3) Is Machine Learning a part of AI or is AI a part of Machine learning? Explain their relationship using a hierarchy diagram in your notebook.

Ans: Machine Learning is a part (subset) of Artifical Intelligence. AI is the broader field that focuses on making machine intelligent. Machine Learning is one of the techniques used to achieve AI.

Within Machine learning, there is another subset called Deep Learning.

Hierarchy Diagram -



- AI is the biggest field.
- Machine Learning is a subset of AI that enables machines to learn from data.
- Deep Learning is a subset of Machine learning that uses neural networks to solve complex problems.

4) Define Machine Learning.
Explain the three main types of Machine learning with one example each:
- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Ans: Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn from data and improve their performance without being explicitly programmed for every task.

PAGE No.
DATE / /

a) Supervised Learning -
Supervised Learning is a type of Machine Learning in which the model is trained using labeled data. This means the input data and the correct output (label) are already known. The model learns the relationship between the input and output and predicts the output for new data.
Example -
Email Spam Detection
- Input : Email
- Output : Spam or Not Spam

b) Unsupervised Learning -
Unsupervised Learning is a type of Machine learning in which the model is trained ~~wi~~ using unlabeled data. Since there are no labels, the model tries to find hidden patterns or groups in the data.
Example -
Customer Segmentation
A shopping website groups customers based on their buying behaviour without knowing the customer categories beforehand.

c) Reinforcement Learning -
Reinforcement learning is a type of Machine Learning in which an agent learns by interacting with its environment. It receives rewards for correct actions and ~~pa~~ penalties for wrong actions. Over time, it learns the best strategy to maximize rewards.
Example -
A robot learning to walk or an AI learning to play chess by repeatedly playing games and improving from rewards & penalties.

DATE / /

5) Differentiate clearly between:
- Supervised Learning
- Unsupervised Learning

Based on:
- Data requirement
- Output
- Real-world applications

Ans:

a) Based on Data Requirement

Supervised Learning requires labeled data, where the correct output is already available for each input.

Unsupervised Learning uses unlabeled data, where only the input data is available, and the model must discover patterns on its own.

b) Based on Output

In Supervised Learning, the output is a prediction or classification because the model learns from known labels.

In Unsupervised Learning, the output is clusters, since there are no predefined labels.

c) Based on Real-World Applications

Supervised Learning is used in applications such as:
- Email Spam Detection
- House Pric ~~P~~ Prediction
- Disease Prediction
- Face Recognition

Unsupervised Learning is used in application such as:
- Customer segmentation

6) What is Reinforcement Learning?

Ans: Reinforcement Learning is a type of Machine Learning in which an agent learns by interacting with its environment. The agent takes actions, receives rewards for correct actions and penalties for wrong actions, and gradually learns the best ~~stat~~ strategy to maximize the total reward.
Unlike supervised learning, Reinforcement Learning does not require labeled data. Instead, it learns through trial and error.

Key Components of Reinforcement learning
- Agent : The learner or decision-maker ~~eg.~~
- Environment : The world in which agent operates
- Action : The decision taken by the agent.
- Reward : Positive or negative feedback received after an action.
- Goal : Learn the best sequence of actions to maximize reward

7) Define ~~8~~ Deep Learning.
How is Deep Learning different from traditional Machine Learning?

Ans: Deep Learning is a subset of Machine Learning that uses Artificial Neural Networks (ANNs) with multiple layers to automatically learns complex patterns from large amounts of data.
In simple words:
Deep Learning is a type of Machine Learning that learns automatically from large amounts of data using neural networks.

Traditional Machine Learning often requires manual feature selection. The programmer decides which features are important for the model.

Deep Learning automatically learns the important features from the data using neural networks, so less manual feature engineering is needed.

Traditional Machine Learning works well with small to medium-sized datasets, whereas Deep Learning performs best with large datasets.

Deep Learning also requires more computing power (GPUs) compared to traditional Machine learning.

8) What is Data Science?
Explain how Data Science is different from Machine Learning.

Ans: Data Science is the process of collecting, cleaning, analyzing, and interpreting data to extract useful insights and help in decision-making.

It combines statistics, programming, data visualization, & Machine learning to solve real-world problems.

Data Science is a broad field that includes the complete process of working with data - from collecting and cleaning data to ~~ana~~ analyzing it and presenting insights.

Machine Learning is a subset of Data Science that focuses only on building models that learn from data and make predictions.

9) Explain the complete Data Science lifecycle :

Ans

- Data Collection -
  The first step is to collect data from different sources such as databases, websites, sensors, APIs, or CSV files.

- Data Cleaning -
  The collected data may contain missing values, duplicate records, or incorrect information. In this step, we ~~to~~ remove errors and prepare the data for analysis.

- Data Analysis -
  The cleaned data is analyzed to discover useful patterns, trends, and relationships.
  Visualization tools like graphs and charts are often used.

- Model Building -
  A suitable Machine Learning algorithm is selected and trained using the prepared data.
  The model learns patterns and becomes capable of making predictions.

- Deployment -
  After testing, the trained model is deployed so that real users or applications can use it. The deployed model continues to make predictions on new data.

10) What is Generative AI (Gen AI)?
How it is different from traditional predictive Machine Learning modules?

PAGE No.
DATE / /

Ans: Generative AI (Gen AI) is a type of Artificial Intelligence that can create new content, such as text, images, music, videos, or code, based on the patterns it has learned from existing data.

Traditional Machine Learning predicts or classifies existing data.

It answers the question:

" What is the correct output?"

Generative AI, on the other hand, creates something new.

" Can I create something new based on what I have learned?"

Examples

Traditional Machine Learning:

- Predict whether an email is spam.

Generative AI:

- Write a complete email from a short prompt